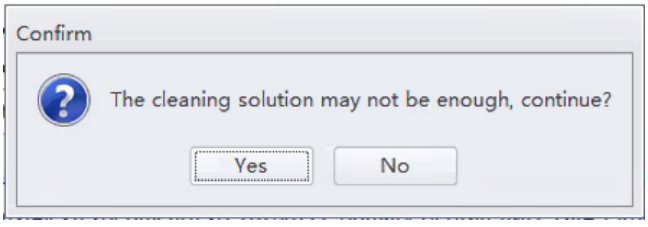
Cytoflex PLATE Mode: General Use, Setup and Shutdown SOP 10/07/2026

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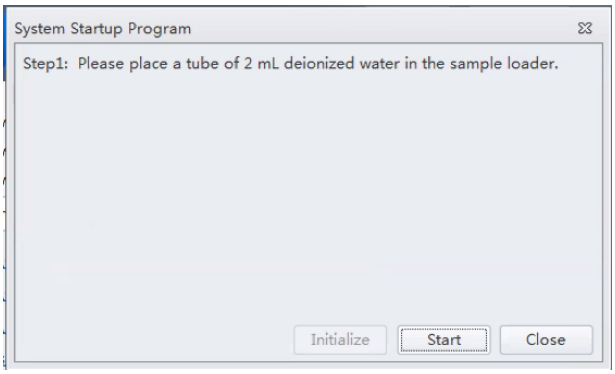
Cytoflex Operating Instructions

- Turn on Cytoflex main power (toggle switch, lower left rear of instrument).
- Turn on the computer's main power.
- Launch the Cytoflex controller software package, CytExpert and ensure that the software connects to the cytometer.
- Check system fluid levels:
    - Check sheath fluid supply level, refill the sheath tank to full with the provided DI water.
        i. If needed fill the Cytoflex sheath boxes with MilliQ water from room 4209
    - Check the waste tank level, if more than half full, remove the lid and empty the tank's contents into the laboratory sink, then add 500mL of bleach to the empty tank and reinstall the tank at the cytometer.
    - Check cleaning solution
        i. Open Cytoflex top lid
        ii. Remove side panel
            1. Loosen both captured thumb screws at the upper corners of side panel
        iii. Check the polypropylene cleaning bottle and refill with 30% Contrad if less than half full.
- In CytExpert: Verify that the Sheath and Waste tank level indicators in the lower right corner are green, indicating that the fluidics levels and system are normal.
- **Fluidic Startup:**
    a. Navigate to Cytometer -> Initialize, allow to finish
    b. Navigate to Cytometer -> System Startup Program
    c. Select "Yes" in the popup window. (This was checked in the fluid level check section.)



    d. Please put at least 2mL of DI water into a FACS Tube and load onto the Sample Station.

e. Press Start in the Startup Program Window.

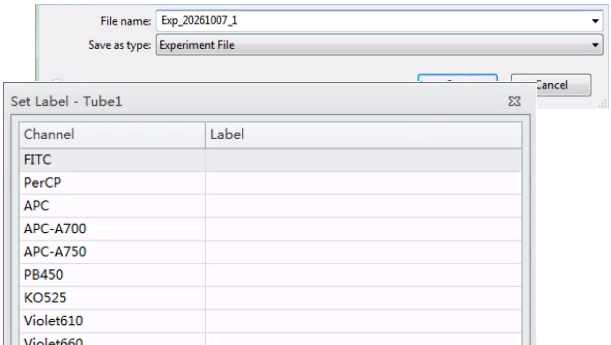


f. Startup Program takes ~6mins

- System QC:
  a. Obtain the current box of Cytoflex QC Beads (Star on box lid) from the refrigerator door.
  b. Vortex the QC bead stock vial vigorously right side up and upside down.
  c. Add drops of the QC bead reagent to the current QC sample FACS Tube until ~400uL total volume (weld line on FACS tube).
     - Note: If no tube is present, use a brand new FACSTube and add drops of the QC bead reagent to the weld line(~400uL).
  d. Vortex the QC bead sample and load onto the sample port
     - If the sample port is not accessible, initialize the system (Cytometer -> Initialize).
     - Navigate to QC/Standardization -> Start QC/Standardization
  e. Make sure the QC radio button is selected
     - Do NOT use Standarsization
  f. Select the proper bead lot from the Lot No. drop down.
     - Located on the QC bottle.
  g. Click “Start”.
  h. Once finished, review the QC Pass/Fail result
     - If Pass: Navigate to File -> Close QC Standardization
     - If Fail, perform a system clean and then remake and retry QC.
  i. Navigate File-> Close QC/Standardization

Experiment Setup (If using a pregenerated template move to section below)

- Click “New Experiment” button in CytExpert popup window.
- Navigate to C:\Users\Bridget\Documents\CytExpert Data
- Select your lab folder (or create new if not present)
- Select your personally folder (or create new if not present)
- Put experiment name into File name field and click “Save”

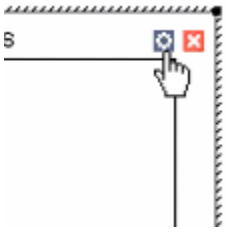


- Click “Acq Setting…”
- Setup Width Parameter
    - Click the “Width” tab
    - In Channel drop down select SSC

- Annotate each parameter with marker used:
    - Click Settings Menu->Set Label
    - Input experiment specific labels for each parameter
    - Press “Apply to..”
    - Make sure “Select All” is selected
    - Press “OK”
    - Click “Close:

- Create experiment worksheet
    - Create the following Bivariate dot plot
        - FSC-A X SSC-A (morphology)
        - SSC-H X SSC-W (Doublet discrimination)
        - live/dead plot (Every experiment should have a live/dead dye or marker to distinguish live cells from dead!– Sadly, FSC-A X SSC-A plot is NOT enough!)

    - Now create a bivariate/contour dot plot or histogram for each remaining color in your panel. You need to have EVERY color represented by at least one plot to properly set up the system. In almost all cases you will want to use the “-A” or area parameter for each channel.
    - Click the “Population Hierarchy” button to bring it into view.
    - Finally, set the SSC axis to log.
        - Click the “cog” button in upper left of FSCxSSC plot
        - In the popup window click the “Log” radio button for SSC.



Acquisition Setup (Detector Gains)

- **“Find your cells” by establishing correct Forward Scatter (FSC) and Side Scatter (SSC) gains for your cell type.**
    - **Load Unstained sample onto the instrument**
    - **Confirm the instrument is set to a low flow rate**
    - **If needed press “Initialize” on the left side panel.**
    - **Press “Run” to start sample acquisition.**
    - **Using the FSCxSSC plot adjust the FSC gain until your cells are on scale and in the optimal range.**
        - **FSC Guidelines:**
            - **All cells on scale and @ or above 10,000(first tick mark) on FSC.**
        - **SSC Guidelines:**
            - **All cells on scale and center of mass between 10^4 - 10^6**

- **Calibrate Detector gains by running full-stained sample/s to represent the brightest possible signal level for each marker in the assay.**
    - **Load full-stained sample onto system**
    - **If needed press “Initialize” on the left side panel.**
    - **Press “Run” to start sample acquisition.**
    - **Using the parameter dot plots or histograms, for each fluorescent marker, adjust its detector gain until all singlet cells are on scale and in the linear range (less than 10^6.5).**

- **Run all single color controls to ensure each is on scale, in the linear range and brightest in its own channel (e.g FITC is brightest in FITC).**
  Note: As long as all single color control signals are on scale and in the linear range for all detectors this step does not affect cytometer data quality but does help with data display and helps to take maximum advantage of some automation features in Flowjo and other analysis packages.
    - **Load the first single color control onto the system**
    - **If needed press “Initialize” on the left side panel.**
    - **Press “Run” to start sample acquisition.**
    - **Check that signals on all detectors are on scale and in the linear range**
    - **Check that the current marker is brightest in its own channel.**
    - ***Note: If a given marker is not brightest in its own channel, either increase the gain of the primary detector or decrease the gain of the brighter secondary detector. It is always better to increase the primary detector\'s gain instead of decreasing the secondary detector’s gain, but be careful that you do not push your full stained sample’s events/cells above 10^6.5 with this increase!',***
    - **When done, click “Stop”.**
    - **Repeat steps for all single color controls.**

- **Setup the desired events to record on the left hand panel.**
    - **This sets the default for all tubes but can be tube/well specific later if needed.**
    - **We recommend at least 10,000 events but for rare event analysis you will need much more. Speak to the flowcore staff to ensure you record enough events.**

- **Experiment and Acquisition setup is complete**

Plate Acquisition Mode:

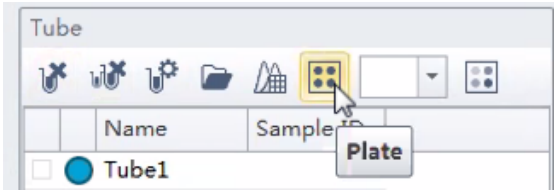
- **Select the plate sample injection mode in software:**
    - Navigate to Cytometer -> Sample Injection Mode -> Choose one of the options
        - Choose Plate Loader Injection Mode for 96 well plate based sample input
- Turn OFF Cytoflex main power (toggle switch, lower left rear of instrument).
- Allow to sit for 5 seconds.
- Turn ON Cytoflex main power (toggle switch, lower left rear of instrument).
- Press OK on software popup:



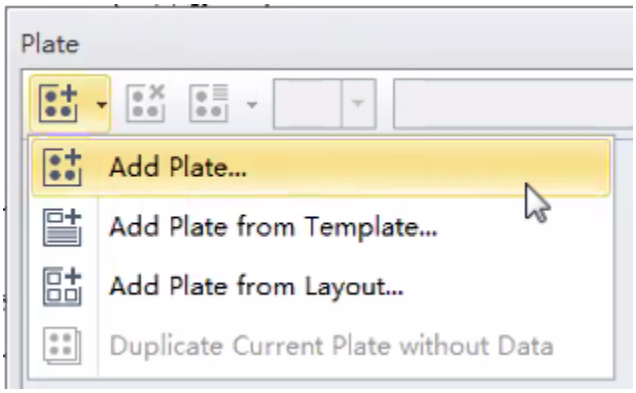
- Confirm software reports Plate Loader. 

- Set the instrument hardware configuration to plate mode.
    a. Look to the left and upwards from the sample probe to locate the tube/plate toggle.
    b. For plate mode the switch should be as shown —> 
- On the left hand panel, select the "Volume to Record" check box.
- Input desired sample uptake volume, **make sure to account for 30ul plate reader dead volume.**
    a. **Min volume: 45ul/well**
    b. **Max volume: 250ul/well**

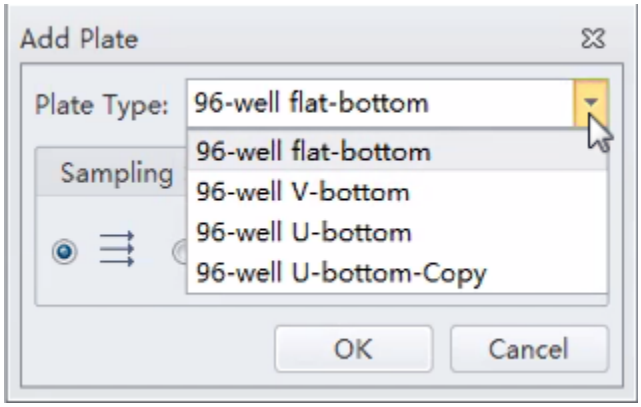
- Click the “Plate” button from the sample section in software.



- Click “Add Plate” in the popup window.



- Select your plate type from the drop down menu in the popup window.

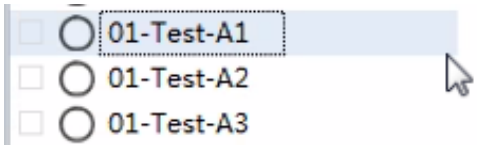


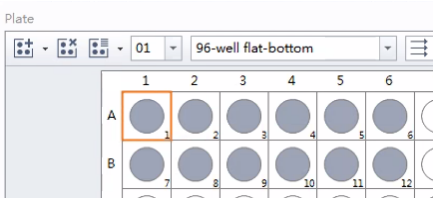
- Press “OK”
- Setup the plate map:
    a. Either single click or click and drag to select sample wells.
    b. Press the “Set As Sample Wells…” button.



    c. In the popup finalize the well prefix and settings for your plate.
        - Global settings were determined during acquisition setup, but ok to establish tube specific settings for stopping rules, meta data ect.. (fluorescent gains must be consistent between samples).
        - Press the “OK” button.

d. Confirm that your desired first well is highlighted orange
    - It is important to select the first desired well in the sample list to ensure all wells are recorded.

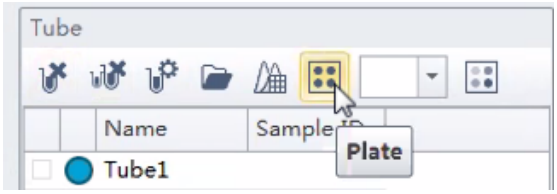




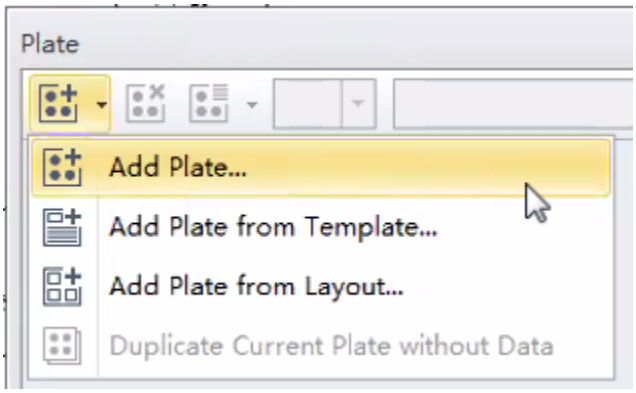
- Click the “Eject” button in the left hand panel to open the Cytoflex plate drawer.
- **Remove your plate lid** and load into the Cytoflex.
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- Click the “Load” button in the left hand panel to load the plate into Cytoflex.
- Click “Auto Record” to begin the plate run.
- Repeat steps for each additional plate.

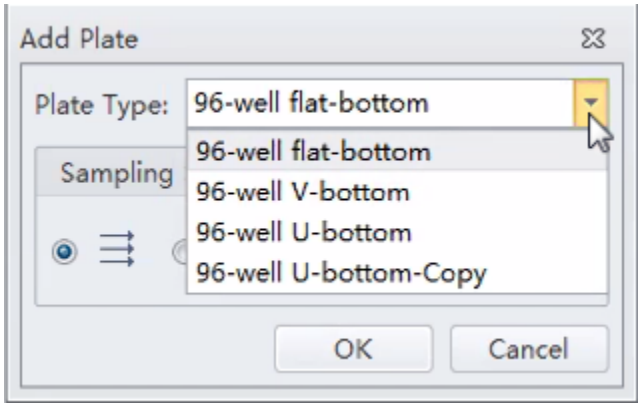
- REQUIRED: Plate mode additional cleaning:
    a. Click the “Fast” sample flow rate radio button.
    b. Prepare the cleaning plate
        - Fill wells A1-A6 completely with ~10% bleach
        - Fill wells A7-A12 completely with DI water.
    c. Click the “Plate” button from the sample section in software.



d. Click “Add Plate” in the popup window.



e. Select your plate type from the drop down menu in the popup window.



f. Press “OK”
g. Select wells A1-A6
h. Press the “Set As Sample Wells…” button.



i. Prefix should be set to “Bleach”
j. Navigate to the Stopping Rules tab.
k. Put 250ul into the “Volume to Record” field.
l. Press “OK”
m. Select wells A7-A12

n. Press the “Set As Sample Wells…” button.



o. Prefix should be set to “DI Water”
p. Navigate to the Stopping Rules tab.
q. Put 250ul into the “Volume to Record” field.
r. Press “OK”
s. Confirm that A1 is highlighted in the sample list.
t. Click the “Eject” button in the left hand panel to open the Cytoflex plate drawer.
u. **Remove the cleaning plate lid** and load into the Cytoflex.
v. Click the “Load” button in the left hand panel to load the plate into Cytoflex.
w. Click “Auto Record” to begin the plate run.

- Return the system to tube mode.
  a. **Select the tube sample injection mode in software:**
     - Navigate to Cytometer -> Sample Injection Mode -> Choose Semi-Automatic for tube based sample input
  b. Confirm instrument hardware is configured for tube mode.
     - Look to the left and upwards from the sample probe to locate the tube/plate toggle.
     - For tube mode the switch should be as shown —>



Shutdown Instrument:

- Click the “Fast” sample flow rate radio button.
- **Click the “Next Tube” button to add a new tube in software and label “Clean”.**
- Prepare a 3ml 10% bleach tube and load onto the system.
- Run the 10% bleach sample for 5 min.
- Select Cytometer->Daily Clean
- Prepare a 3ml tube of 10% Contrad and load onto the Cytoflex
- Click “Run” in the Daily Clean wizard.
- Allow to finish
- Prepare a 3ml tube of DI water and load onto the Cytoflex.
- Click “Run” in the Daily Clean wizard.
- Turn off Cytoflex main power.
- Close CytExpert software.
- Go to the last page of SOP for data transfer instructions.

Transferring data files:

Preferred Option 1: Using ucdavis.help@gmail.com communal flow core email address via Google Drive sharing:

- Using ucdavis.help@gmail.com Google Drive navigate to "Flow Core User Data Transfers\\(DAVIS) BC Cytoflex - Tupper".
- Drag and drop your exported data folder (C:\Users\Bridget\Documents\CytExpert Data\”Your Lab Name”\”Your Name”) to the Google Drive
- Right click the newly imported data folder and Select Share from the context menu, then click Share again.
- Under the "General access" section, click the dropdown menu (which defaults to "Restricted") and change it to Anyone with the link.
- Click the Copy link button in the bottom left corner of the dialog box, then click Done.
- Create and send a new email with your UCD email as recipient and paste the copied link into its body.

Option 2: Your own USB external drive/data stick

- Plug in external drive
- Navigate to C:\Users\Bridget\Documents\CytExpert Data\”Your Lab Name”\”Your Name”
- Transfer data to external drive.